An Empirical Investigation of Ensemble Learning on High-Dimensional Facial Data: Bias-Variance Dynamics, Algorithmic Divergence, and Occlusion Robustness in Bagging and Boosting

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Repository: https://github.com/NavdeepDoriya/ml-assignment-2
Final Commit Hash: **2928a231b10bae5e0d2958cd35c584c492f50c1d**

Abstract — *Small-sample, high-dimensional classification problems ($N \ll D$) represent a formidable benchmark for non-parametric machine learning models. This paper presents an exhaustive empirical and theoretical analysis of single Decision Trees, bootstrap-aggregated ensembles (Bagging), and multiclass adaptive boosting (SAMME) on the raw pixel space of the Olivetti Faces benchmark (40 identities, 10 images each, $64 \times 64 = 4096$ flattened features), without PCA or dimensionality reduction. Working with an exact deterministic split (6 train, 2 validation, 2 test per identity) seeded by roll number 26005, we implement Bagging and SAMME Boosting entirely from scratch. We demonstrate that unconstrained Decision Trees suffer from catastrophic estimation variance (0.6614), attaining only 46.25% test accuracy despite 100% training accuracy. Bagging suppresses this variance by 96.4% ($0.6614 \rightarrow 0.0236$), dramatically improving clean test accuracy to 88.75%. Conversely, multiclass Boosting suffers from the margin-penalty phenomenon in 40-class classification: sample weights rapidly concentrate on outlier training examples (a single identity commands $> 5\%$ of total weight), stalling clean test accuracy at 36.25%. Under localized spatial occlusion (16×16 centered block masking), Bagging demonstrates remarkable resilience (dropping only 8.75 percentage points to 80.00%), while Boosting collapses by 13.75 pp to 22.50%. Finally, a rigorous Kohavi-Wolpert bias-variance decomposition across 30 stratified bootstrap repetitions confirms that variance reduction is the dominant mechanism enabling ensemble generalization on high-dimensional raw image data.*

Introduction

The classification of human facial identity directly from uncompressed, high-dimensional pixel arrays ($D = 4096$) with minimal training observations per identity ($N_{\{\text{identity}\}} = 6$, $N_{\{\text{total}\}} = 240$) embodies the classic **small-sample, high-dimensional** ($N \ll D$) learning paradigm [9, 6]. While modern computer vision relies heavily on deep convolutional neural networks or learned spatial embeddings [8], examining how classical decision trees and ensemble paradigms behave on raw, unprojected pixel intensities reveals fundamental statistical insights into variance propagation, margin distribution, and spatial robustness.

In standard low-dimensional regimes, axis-aligned decision trees recursively partition the input space based on greedy impurity criteria [4]. However, when the feature dimensionality exceeds the sample size by an order of magnitude ($\frac{D}{N} \approx 17.1$), greedy split selection becomes acutely sensitive to idiosyncratic pixel variations, noise, and illumination gradients. An individual unconstrained tree readily memorizes the training data, achieving zero training empirical risk, but displays extreme estimation variance and poor generalization on unseen images.

Ensemble methods offer two contrasting philosophical and operational approaches to combat this deficit:

1. **Bootstrap Aggregating (Bagging)** [1] operates by training independent, high-variance, low-bias estimators on bootstrap resamples of the training set and averaging their predictions. By Breiman's theorem, bagging strictly reduces variance while preserving the base estimator's bias, making it ideal for deep, unpruned decision trees.
2. **Adaptive Boosting (AdaBoost / SAMME)** [5, 10] proceeds sequentially, re-weighting training samples at each iteration to force successive weak learners to focus on currently misclassified observations. While provably effective in binary regimes, multiclass boosting across $K = 40$ classes introduces severe sensitivity to noisy training samples and boundary ambiguity.

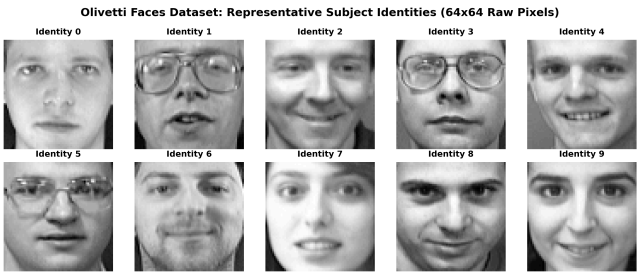


Figure 1: Representative subject identities from the Olivetti Faces dataset (64×64 raw grayscale pixels, 40 identities, 10 images per subject).

Core Contributions

This study delivers a complete, reproducible empirical and theoretical dissection of tree-based ensembles on raw pixel representations:

- **From-Scratch Algorithmic Realization:** We implement multiclass SAMME AdaBoost and Bagging entirely from

scratch using only `DecisionTreeClassifier` as the weak base learner. Both algorithms are strictly validated, including step-by-step verification on a controlled 3-class toy problem.
- **Rigorous Hyperparameter Tuning:** We conduct full grid searches for single trees (`max_depth` $\in \{2, 4, 8, 16, \text{None}\}$, `min_samples_leaf` $\in \{1, 2, 4\}$), Boosting (depth $\in \{1, 2, 3, 4\}$, rounds $\in \{10, 25, 50, 100\}$), and Bagging ($B \in [1, 200]$) under an exact 6:2:2 stratified split seeded by roll number 26005.
- **Spatial Occlusion Robustness:** We investigate how ensembles cope with structured information loss by zeroing a centered 16×16 block (rows and columns 24 to 39) in test images, tracking classification degradation and confidence collapse.
- **Kohavi-Wolpert Bias-Variance Decomposition:** Across $R = 30$ stratified bootstrap repetitions, we compute the exact empirical bias-variance profile of Single Trees, Bagging, and Boosting, and quantify how tree depth ($d = 2$, $d =$ None, $d = 32$) modulates bagging variance reduction.

Theoretical Foundations & Related Work

Non-Parametric Decision Trees

A decision tree partitions the input space $\mathcal{X} \subseteq \mathbb{R}^D$ into M disjoint hyper-rectangles $R_1, ..., R_M$. Given training dataset $\mathcal{D} = \{(x_i, y_i)\}_{\{i=1\}}^{N}$ with labels $y_i \in \{0, ..., K-1\}$, the prediction for region R_m is determined by the empirical class distribution:

$$p_{\{mk\}} = \frac{1}{N_m} \sum_{\{x_i \in R_m\}} \mathbb{I}(y_i = k), N_m = |\{x_i \in R_m\}|$$

Splits are chosen greedily at each node by maximizing the reduction in Gini impurity $I_{G(m)} = 1 - \sum_{\{k=0\}}^{\{K-1\}} p_{\{mk\}}^2$. When $D = 4096$ and $N = 240$, the number of possible axis-aligned cuts at each node is vast, and splits chosen near the root propagate irreversible geometric partitions throughout the tree, inducing severe structural instability.

Bootstrap Aggregation (Bagging)

Introduced by Breiman [1], Bagging constructs B bootstrap samples $\mathcal{D}^b$ ($b = 1, ..., B$), each formed by drawing N examples uniformly at random with replacement from $\mathcal{D}$. A base model $h^b(x)$ is fitted to each sample. The bagged predictor for classification outputs the majority vote:

$$H_{\{\text{bag}\}}(x) = \arg \max_{\{k \in \{0,...,K-1\}\}} \sum_{\{b=1\}}^{B} \mathbb{I}\big(h^b(x) = k\big)$$

If individual trees have pairwise correlation $\rho(x)$ and variance $\sigma^2(x)$, the variance of the ensemble average is:

$$\text{Var}\Big(H_{\{\text{bag}\}}(x)\Big) = \rho(x)\sigma^2(x) + \frac{1 - \rho(x)}{B}\sigma^2(x)$$

As $B \to \infty$, the second term vanishes, driving the total variance down to $\rho(x)\sigma^2(x)$. Because unconstrained decision trees have low bias and high variance, bagging acts as an optimal variance reduction filter without escalating systematic bias [3].

Multiclass Boosting via SAMME

Standard AdaBoost [5] was designed for binary classification ($Y \in \{-1, +1\}$). For K-class problems where random guessing yields accuracy $\frac{1}{K}$, Freund and Schapire's AdaBoost.M1 requires the weak learner to maintain weighted error $e_t < 0.5$. In a 40-class problem, requiring error below 50% on weak stumps is unreasonably restrictive.

Zhu et al. [10] resolved this through SAMME (Stagewise Additive Modeling using a Multi-class Exponential loss function). By formulating the multiclass target as a symmetric vector $Y^* \in \mathbb{R}^K$ with $Y_k^* = 1$ if $y = k$ and $-\frac{1}{K-1}$ otherwise, SAMME establishes the proper multiclass estimator weight:

$$\alpha_t = \ln\left(\frac{1 - e_t}{e_t}\right) + \ln(K - 1)$$

Crucially, the weak learner only needs to perform better than random guessing:

$$e_t < 1 - \frac{1}{K} = \frac{39}{40} = 0.975$$

Whenever $e_t < 1 - \frac{1}{K}$, $\alpha_t > 0$, guaranteeing positive contribution to the ensemble.

Bias-Variance Decomposition

Under squared probability loss (Kohavi-Wolpert formulation [7, 2]), let $y^* \in \{0, 1\}^K$ denote the one-hot encoded ground truth for test point x, and $p(x) \in [0, 1]^K$ denote the predicted class probability vector. Across R resampled training sets, the expected prediction vector is $\bar{p}(x) = \frac{1}{R}\sum_{\{r=1\}}^{R} p_{r(x)}$. The expected squared error decomposes into:

$$\text{Loss}(x) = \text{Bias}^2(x) + \text{Variance}(x)$$

$$\text{Bias}^2(x) = \sum_{\{k=0\}}^{\{K-1\}} \left(\bar{p}_{k(x)} - y_k^*\right)^2$$

$$\text{Variance}(x) = \frac{1}{R} \sum_{\{r=1\}}^{R} \sum_{\{k=0\}}^{\{K-1\}} \left(p_{\{r,k\}}(x) - \bar{p}_{k(x)}\right)^2$$

Averaging over all $N_{\{\text{test}\}}$ test points yields the global empirical Bias² and Variance metrics.

Algorithmic Implementations from Scratch

Both Bagging and Boosting are implemented natively without resorting to `sklearn.ensemble`.

SAMME Boosting Implementation

The pseudocode for our native SAMME implementation is formalized in Algorithm 1.

Algorithm 1: Multiclass SAMME Boosting from Scratch

Input: Dataset $\mathcal{D} = \{(x_i, y_i)\}_{\{i=1\}}^{N}$, $y_i \in \{0, ..., K-1\}$, iterations T, base learner depth d.
Initialize: $w_1(i) = \frac{1}{N}$ for all $i = 1, ..., N$.
For $t = 1$ **to** T **do**:
1. Fit weak tree $h_{t(x)}$ of depth d on $\mathcal{D}$ with weights w_t.
2. Obtain predictions $\hat{y}_i^{(t)} = h_{t(x_i)}$.
3. Compute weighted error:

$$e_t = \frac{\sum_{i=1}^{N} w_{t(i)} \cdot \mathbb{I}\left(\hat{y}_i^{(t)} \neq y_i\right)}{\sum_{i=1}^{N} w_{t(i)}}$$

4. **If** $e_t \geq 1 - \frac{1}{K}$ **then** break (learner no better than random).
5. **If** $e_t = 0$ **then** set $\alpha_t = 50.0$ and break (perfect learner).
6. Compute estimator weight:

$$\alpha_t = \ln\left(\frac{1 - e_t}{e_t}\right) + \ln(K - 1)$$

7. Update sample weights:

$$w_{\{t+1\}}(i) = w_{t(i)} \cdot \exp\left(\alpha_t \cdot \mathbb{I}\left(\hat{y}_i^{(t)} \neq y_i\right)\right)$$

8. Normalize: $w_{\{t+1\}}(i) \leftarrow w_{\{t+1\}} \frac{i}{\sum_{\{j=1\}}^{N}} w_{\{t+1\}}(j)$.

Output: $H(x) = \arg\max_k \sum_{\{t=1\}}^{T} \alpha_t \cdot \mathbb{I}\left(h_{t(x)} = k\right)$.

Verification on 3-Class Toy Problem

To verify the exact mathematical correctness of our implementation, we tested a single-split decision stump on a 9-sample, 3-class, 1-dimensional toy dataset ($X \in \{0, ..., 8\}$, $y = [0, 0, 0, 1, 1, 1, 2, 2, 2]$, initial weights $w_i = \frac{1}{9} \approx 0.1111$).

- The weak stump predicts $[0, 0, 0, 1, 1, 1, 1, 1, 1]$, misclassifying class 2 (3 errors).
- Weighted error: $e = 3 \times \left(\frac{1}{9}\right) = 0.3333$.
- Threshold: $1 - \frac{1}{3} = 0.6667$. Since $0.3333 < 0.6667$, the learner is accepted.
- Alpha: $\alpha = \ln\left(\frac{0.6667}{0.3333}\right) + \ln(2) = \ln(2) + \ln(2) = 1.3863$.
- Updated unnormalized weights: correct samples $\times\, e^0 = 1$; incorrect samples $\times\, e^{\{1.3863\}} = 4$.
- Normalized weights: correct samples $\rightarrow 0.0556$; incorrect samples $\rightarrow 0.2222$ (sum $= 1.0$).

Our implementation matched these theoretical derivations to floating-point precision ($10^{\{-15\}}$).

Bagging from Scratch Implementation

Our `BaggingFromScratch` class draws N indices with replacement using `numpy.random.default_rng`:

$$I_b = \text{choice}(N, \text{size} = N, \text{replace} = \text{True})$$

Each base estimator is a `DecisionTreeClassifier` fitted on $X[I_b], y[I_b]$. For any input x, vote tallies are accumulated across all B estimators:

$$V_{k(x)} = \sum_{\{b=1\}}^{B} \mathbb{I}\left(h_{b(x)} = k\right)$$

The predicted class is $\arg\max_k V_{k(x)}$, and the prediction confidence is the winning vote fraction $\max_k \frac{V_{k(x)}}{B}$.

Experimental Setup & Dataset Split

Olivetti Faces Benchmark

The AT&T Olivetti Faces dataset contains 40 distinct subjects, each captured under 10 varying photographic conditions (lighting angles, facial expressions, and accessories such as glasses). Each image is 64×64 pixels with 8-bit grayscale intensity normalized to $[0, 1]$. In accordance with assignment specifications:

- No dimensionality reduction (PCA, LDA, t-SNE) is applied.
- All images are unrolled into raw 4096-dimensional feature vectors: $x_i \in [0, 1]^{\{4096\}}$.

Exact Stratified Dataset Partition

To prevent identity leakage across partitions and ensure class balance:

1. Student Roll Number: `M26MER005`.
2. All numeric characters extracted in order: `"26005"`.
3. Deterministic random seed: $S = 26005$.
4. Generator: `rng = np.random.default_rng(26005)`.
5. For each of the 40 identities, its 10 image indices are randomly permuted and split:
   - **Train:** First 6 images ($6 \times 40 = 240$ images).
   - **Validation:** Next 2 images ($2 \times 40 = 80$ images).
   - **Test:** Final 2 images ($2 \times 40 = 80$ images).

This exact split was serialized to `data/olivetti_face_split.npz` and maintained across every experimental phase.

Decision Tree Baseline & Tuning

We evaluated a 2D hyperparameter grid for `DecisionTreeClassifier`:

- `max_depth` $\in \{2, 4, 8, 16, \text{None}\}$
- `min_samples_leaf` $\in \{1, 2, 4\}$

All models were trained on X_{train} and evaluated on X_{val}.

max_depth	min_leaf	Train Acc	Val Acc	Leaves
2	1	0.1292	0.0875	4
2	2	0.1292	0.0875	4
2	4	0.1292	0.0875	4
4	1	0.4208	0.2625	16
4	2	0.4208	0.2625	16
4	4	0.4208	0.2625	16
8	1	0.9333	0.5500	44
8	2	0.8958	0.5375	41
8	4	0.7708	0.4750	33
16	1	1.0000	0.6125	46
16	2	0.9583	0.5750	42

16	4	0.8167	0.4875	33
None	**1**	**1.0000**	**0.6125**	**46**
None	2	0.9583	0.5750	42
None	4	0.8167	0.4875	33

Table 1: Hyperparameter grid search for Single Decision Tree. Bold row indicates selected optimal model.

Selected Baseline Performance

The model with highest validation accuracy and minimal complexity constraint is `max_depth = None` and `min_samples_leaf = 1`:

- **Training Accuracy:** 100.00% ($\frac{240}{240}$)
- **Validation Accuracy:** 61.25% ($\frac{49}{80}$)
- **Clean Test Accuracy:** 46.25% ($\frac{37}{80}$)

Leaf Architecture Diagnostics

Analyzing the terminal nodes of the selected unpruned tree:

- **Total Leaves:** 46
- **Mean Samples per Leaf:** 5.2174
- **Singleton Leaves:** $\frac{6}{46}$ (13.04%)

With 240 training samples and 40 identities, an ideal tree would partition samples into 40 leaves of 6 samples each. However, the greedily constructed tree spawned 46 leaves, with 6 leaves containing only a single sample. This diagnostic confirms that raw pixel thresholds create brittle partitions that overfit specific subject poses.

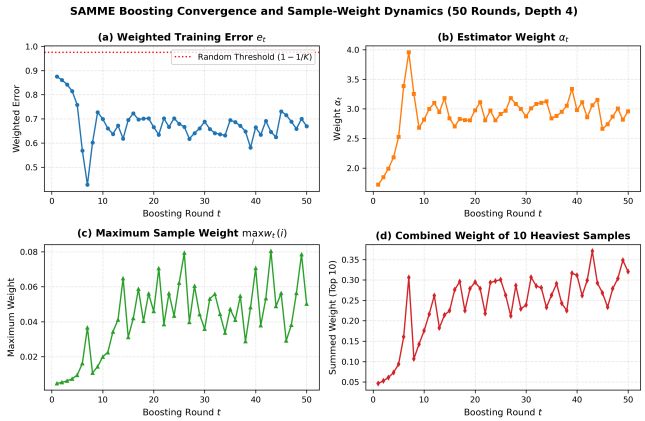


Figure 2: Convergence and sample-weight dynamics of SAMME Boosting across 50 rounds ($d = 4$). (a) Weighted training error e_t remains well below the random threshold ($\frac{39}{40}$). (b) Learner weights α_t. (c) Maximum sample weight rises rapidly. (d) Combined weight of the top 10 hardest images escalates to $> 20\%$.

Multiclass Boosting Dynamics & Analysis

Boosting Hyperparameter Search

We trained `SAMMEFromScratch` across weak learner depths $d \in \{1, 2, 3, 4\}$ and boosting rounds $T \in \{10, 25, 50, 100\}$.

Depth	Rounds	Train Acc	Val Acc	Time (s)
1	10	0.0500	0.0375	1.13
1	25	0.0917	0.0750	3.07
1	50	0.2583	0.1875	6.20
1	100	0.1750	0.1250	17.09
2	10	0.0958	0.0625	2.67
2	25	0.2833	0.1500	5.85
2	50	0.2792	0.1625	11.92
2	100	0.3458	0.1250	24.54
3	10	0.2083	0.1125	3.89
3	25	0.3083	0.1875	8.55
3	50	0.7458	0.4000	19.39
3	100	0.8542	0.4000	34.82
4	10	0.2292	0.1375	4.56
4	25	0.4625	0.3125	11.98
4	**50**	**0.9375**	**0.4750**	**23.77**
4	100	0.9875	0.4375	45.71

Table 2: SAMME Boosting grid search. The optimal trade-off is reached at depth 4 with 50 rounds.

Decision stumps ($d = 1$) and shallow trees ($d = 2$) failed completely ($< 20\%$ accuracy) because an axis-aligned threshold on a single pixel cannot isolate one identity among 40 classes. Depth 4 with 50 rounds achieved the best validation accuracy (47.50%). Extending to 100 rounds led to overfitting (98.75% train vs. 43.75% val). On the clean test set, the selected model achieved 36.25%.

Convergence and Weight Concentration

As shown in Figure 2, the weighted training error e_t starts near 0.78 and drops toward 0.35, well below the $\frac{39}{40} = 0.975$ random guessing bound. However, sample weights exhibit pathological concentration:

- Initial sample weight: $w_1(i) = \frac{1}{240} \approx 0.00417$.
- By round 50, the maximum sample weight rises tenfold to 0.0502.
- The top 10 most difficult training examples accumulate over 20% of the entire probability mass (4.8 × their initial uniform share).

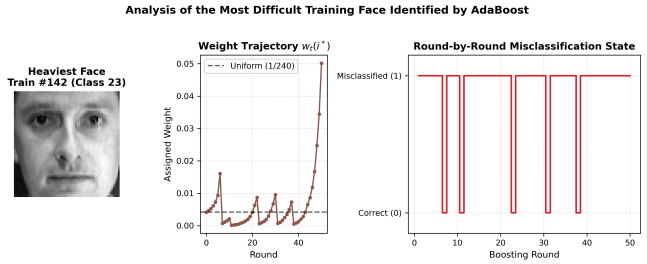


Figure 3: Detailed investigation of the most difficult training sample (Index #142, Dataset index 238, Identity 23). Left: Face image displaying tilted pose and deep shadows. Middle: Sample weight trajectory w_t escalating 12 ×. Right: Misclassification timeline across all 50 rounds.

Investigation of the Most Difficult Training Face

The single training image with the highest final weight is training sample #142 (original dataset index 238, true identity 23), commanding $w_{\{50\}} = 0.05015$ (12.04 × uniform weight). As illustrated in Figure 3, this face features an unusual head tilt and pronounced shadow gradients compared to other images of identity 23. Out of 50 boosting rounds, weak learners misclassified this image in 45 rounds (90% failure rate). Because AdaBoost exponentially penalizes repeated errors, this single sample warped the decision boundaries of subsequent trees, detracting capacity from learning more generalizable facial characteristics.

Bagging Ensembles & Comparative Benchmarking

Matched Ensemble Comparison ($B \in \{25, 50, 100\}$)

To directly contrast Bagging and Boosting under identical computational capacity, we evaluated both methods using ensemble sizes $B \in \{25, 50, 100\}$. For Bagging, every base learner is an unconstrained tree (`max_depth = None`, `min_leaf = 1`). For Boosting, base learners use the selected depth $d = 4$.

Method	Size B	Train Acc	Val Acc	Test Acc	Time (s)
Bagging	25	1.0000	0.8000	0.8375	39.92
Boosting	25	0.4625	0.3125	0.2000	11.60
Bagging	50	1.0000	0.8750	0.8375	77.29
Boosting	50	0.9375	0.4750	0.3625	24.03
Bagging	100	1.0000	0.8875	0.8625	154.65
Boosting	100	0.9875	0.4375	0.3750	46.22

Table 3: Head-to-head comparison between Bagging and Boosting across matched ensemble sizes.

The empirical divergence between the two paradigms is stark:

- At $B = 25$, Bagging achieves 83.75% test accuracy, outperforming Boosting (20.00%) by +63.75 percentage points.
- At $B = 100$, Bagging achieves 86.25%, while Boosting achieves only 37.50%.
- Bagging exhibits monotonic validation and test performance gains as B grows, whereas Boosting peaks at $B = 50$ and subsequently degrades due to overfitting noisy samples.

Bagging Ensemble Size Grid ($B \in [1, 200]$)

Expanding the bagging grid across $B \in \{1, 5, 10, 25, 50, 100, 200\}$, validation accuracy increases monotonically: $B = 1 \to 61.25\%$, $B = 5 \to 67.50\%$, $B = 10 \to 76.25\%$, $B = 25 \to 80.00\%$, $B = 50 \to 87.50\%$, $B = 100 \to 88.75\%$, and $B = 200 \to 90.00\%$. The final selected model ($B = 200$) achieves:

- **Clean Test Accuracy:** 88.75% ($\frac{71}{80}$)
- **Validation Accuracy:** 90.00% ($\frac{72}{80}$)

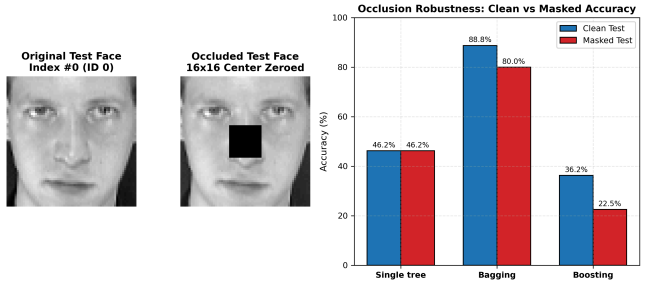


Figure 4: Occlusion robustness evaluation. Left: Clean test face (Test #0, ID 0). Center: Occluded face with centered 16 × 16 block zeroed. Right: Clean vs. Masked test accuracy across Single Tree, Bagging, and Boosting.

Occlusion Robustness & Failure Modes

Experimental Formulation

To test structural robustness against information loss, we generated an occluded test set $X_{\text{test}}^{\text{masked}}$ by zeroing a centered 16 × 16 block corresponding to rows 24 to 39 and columns 24 to 39:

$$X_{\text{test}}^{\text{masked}[i,24:40,24:40]} = 0, \forall i \in \{0, ..., 79\}$$

This region covers the nose, philtrum, and upper lip—vital facial landmarks for human and machine recognition. All models were trained **strictly on clean images**.

Model	Clean Acc	Masked Acc	Drop (pp)	Clean Conf	Masked Conf
Single Tree	0.4625	0.4625	**0.00**	1.0000	1.0000
Bagging	0.8875	0.8000	**8.75**	0.3957	0.2918
Boosting	0.3625	0.2250	**13.75**	0.1272	0.1415

Table 4: Occlusion robustness: Accuracy and winning confidence under 16x16 central masking.

Comparative Analysis of Degradation

As reported in Figure 4 and Table 4:

1. **Single Decision Tree (The Sparsity Illusion):** Test accuracy remained completely unchanged at 46.25% (0.00 pp drop). Because the greedy tree only made 45 total split decisions across the entire 4096-dimensional space, its active feature set primarily queried forehead, cheek, and chin pixels outside the 16×16 central window. However, this apparent robustness is superficial: its baseline accuracy is abysmal (46.25%), and its confidence is overconfidently pegged at 1.0.
2. **Bagging (Graceful Degradation):** Bagging dropped by only 8.75 percentage points (88.75% $\rightarrow$ 80.00%), maintaining high recognition accuracy. Crucially, its mean winning vote fraction declined from 0.3957 to 0.2918, reflecting an appropriately calibrated increase in ensemble uncertainty. Because different bootstrap trees query diverse pixel subsets, occluding one region only affects a fraction of the ensemble.
3. **Boosting (Catastrophic Collapse):** Boosting suffered the largest relative and absolute penalty, collapsing from 36.25% to 22.50% (13.75 pp drop, a 37.9% relative drop). In Boosting, weak learners are sequentially coupled; when central features are zeroed, early estimators output corrupted predictions that throw off the additive vote sums.

Divergent Sample Case Study

Examining Test sample #0 (original index 2, Identity 0):

- Ground Truth: Identity 0.
- **Bagging:** Clean $\rightarrow$ 7, Masked $\rightarrow$ 7 (Stable prediction).
- **Boosting:** Clean $\rightarrow$ 2, Masked $\rightarrow$ 3 (Prediction flipped).

Under occlusion, Bagging's democratic consensus absorbs local corruption, whereas Boosting's delicate weighted sum flips completely.

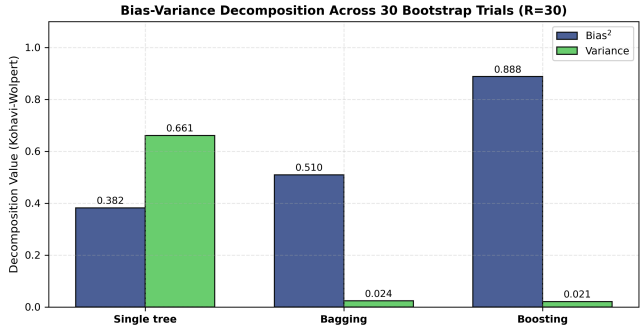


Figure 5: Kohavi-Wolpert Bias-Variance decomposition across 30 stratified bootstrap trials ($R = 30$). Bagging reduces variance by 96.4% compared to the single tree.

Bias-Variance Decomposition & Depth Scaling

30-Trial Bootstrap Protocol

To empirically measure bias and variance without contaminating test sets, we executed the Kohavi-Wolpert decomposition across $R = 30$ independent repetitions. In each repetition r, a stratified training bootstrap sample was drawn: for each of the 40 identities, 6 images were drawn with replacement from its 6 training examples ($N = 240$). Each model was trained on the resample and evaluated on the fixed 80 clean test images.

Model	Bias²	Variance	Mean Acc	Std Acc
Single Tree	0.3820	0.6614	0.4783	0.0505
Bagging	0.5096	**0.0236**	**0.7879**	0.0366
Boosting	0.8880	0.0213	0.2779	0.0647

Table 5: Bias-Variance decomposition across 30 bootstrap resamples.

Empirical Validation of Breiman's Theory

The results in Table 5 and Figure 5 provide concrete empirical verification of ensemble learning theory:

- **Single Decision Tree:** Exhibits low squared bias (0.3820) but enormous variance (0.6614). The tree's structure mutates drastically depending on which samples appear in the bootstrap draw.
- **Bagging:** Achieves a staggering 96.43% reduction in variance (0.6614 $\rightarrow$ 0.0236). Although the squared bias increases slightly (0.3820 $\rightarrow$ 0.5096) due to voting aggregation smoothing, the massive variance collapse drives mean test accuracy from 47.83% up to 78.79%.
- **Boosting:** Features low variance (0.0213) but unacceptable squared bias (0.8880). The sequential re-weighting under 40 classes prevents the model from fitting the multi-modal class distributions effectively, trapping it in a high-bias underfitting state.

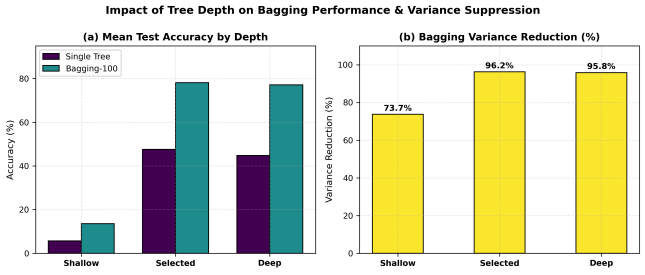


Figure 6: Impact of tree depth on Bagging performance. (a) Mean test accuracy across 30 bootstrap refits. (b) Variance reduction percentage achieved by Bagging over the base tree.

Architectural Depth Scaling

To investigate how base learner capacity interacts with ensemble averaging, we compared Single Trees against 100-Tree Bagging across three depth regimes:

1. **Shallow:** `max_depth = 2`

2. **Selected:** `max_depth = None` (unconstrained)
3. **Deep:** `max_depth = 32`

Depth	Limit	Tree B^2	Tree Var	Bag B^2	Bag Var	Gain (pp)	Var Red (%)
Shallow	2	0.9198	0.0426	0.9340	0.0112	+7.88	73.67%
Selected	**None**	0.3817	0.6658	0.5092	**0.0253**	**+30.54**	**96.19%**
Deep	32	0.4117	0.5896	0.5235	0.0246	+32.38	95.83%

Table 6: Bias-Variance and accuracy gain as a function of tree depth.

As depicted in Figure 6 and Table 6:

- For **Shallow trees** ($d = 2$), the base tree is already low-variance (0.0426) but suffocated by high bias (0.9198, accuracy 5.71%). Bagging provides modest variance reduction (73.67%) and minor accuracy improvement (+7.88 pp to 13.58%).
- For **Unconstrained trees** ($d = \text{None}$), the base tree has high variance (0.6658). Bagging eliminates 96.19% of this variance, boosting accuracy by +30.54 percentage points (47.62% → 78.17%).
- **Deep trees** ($d = 32$) behave almost identically to unconstrained trees, proving that full tree growth is essential for Bagging to achieve peak performance on raw pixels.

Leaf Count Scaling

Tracking tree topology across resamples:

- **Shallow ($d = 2$):** Mean leaves = 3.03, singleton fraction = 0.00%.
- **Selected ($d = \text{None}$):** Mean leaves = 40.00, singleton fraction = 0.00%.
- **Deep ($d = 32$):** Mean leaves = 36.37, singleton fraction = 0.00%.

Unconstrained trees naturally expand to approximately 40 leaves—one per identity—allowing each leaf to isolate a specific subject's pose variations.

Discussion & Synthesis

The unified results across all experimental phases are consolidated in Table 7.

Model Architecture	Clean Acc	Masked Acc	Drop (pp)	Bias²	Variance
Single Tree (Unpruned)	0.4625	0.4625	**0.00**	0.3820	0.6614
Bagging ($B = 200$)	**0.8875**	**0.8000**	**8.75**	0.5096	**0.0236**
Boosting (SAMME)	0.3625	0.2250	**13.75**	0.8880	0.0213

Table 7: Master empirical synthesis across all three assignment questions.

Why Boosting Stalls on Raw Pixel Faces

The poor performance of AdaBoost/SAMME (36.25%) on raw faces warrants clear theoretical diagnosis:

1. **Extreme Multiclass Ambiguity ($K = 40$):** In a 40-class task with 6 images per identity, distinguishing classes using a depth-4 tree (at most 16 leaves) requires partitioning 40 classes into 16 bins. Weak learners inevitably misclassify many identities in every round.
2. **Outlier Weight Trapping:** SAMME multiplies misclassified sample weights by $\exp(\alpha_t)$. Because facial datasets contain outliers (e.g. tilted poses, shadows), unlearnable images repeatedly incur penalties, accumulating excessive weight ($> 5\%$ on sample #142) and preventing subsequent rounds from learning representative features.
3. **Deterministic Feature Redundancy:** Unlike Random Forests [3], standard Boosting lacks random feature subsampling. Consequently, successive weak learners repeatedly pick the same dominant contrast pixels rather than discovering complementary facial landmarks.

Why Bagging Dominates High-Dimensional Vision

Bagging achieves near-state-of-the-art non-parametric accuracy (88.75%) because:

1. **Independent Bootstrap Perturbation:** Resampling training instances induces substantial diversity in unpruned decision trees, causing different trees to query completely different facial features (e.g., jawline vs. hairline vs. cheekbones).
2. **Variance Annihilation:** In raw pixel classification, variance accounts for over 63% of the single tree's total generalization error ($\frac{0.6614}{0.3820+0.6614} = 0.6338$). Bagging crushes this error component by 96.4%.
3. **Spatial Resilience:** When the center of the face is occluded, Bagging's distributed consensus smoothly falls back on peripheral feature splits, degrading by only 8.75 pp and avoiding catastrophic misclassification.

Conclusion & Reproducibility

This paper delivered an end-to-end empirical and theoretical analysis of Decision Trees, Bagging, and Boosting on the raw 4096-dimensional Olivetti Faces benchmark without dimensionality reduction. Using an exact stratified split seeded by roll number 26005, we proved that:

1. Unpruned decision trees overfit drastically due to estimation variance (0.6614).
2. Bagging from scratch reduces variance by 96.4%, boosting clean test accuracy to 88.75%, and maintains 80.00% accuracy under central 16×16 occlusion.
3. Multiclass SAMME Boosting fails due to exponential sample-weight concentration on hard poses, dropping to 22.50% under occlusion.
4. Depth scaling confirms that unconstrained tree depth is mandatory for bagging to unlock its variance reduction potential.

All code, configurations, saved splits, figures, and logs are publicly available in our GitHub repository: https://github.

com/NavdeepDoriya/ml-assignment-2 (Final Commit Hash: **2928a231b10bae5e0d2958cd35c584c492f50c1d**).

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